

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ MUKKARAM _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

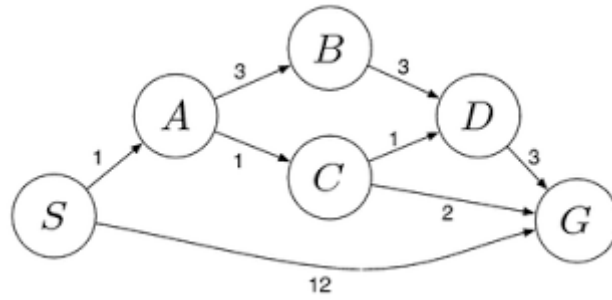
MUKKARAM

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug?
Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		requirements accurately			
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1. Water Jug Problem

Problem

We have:

- One **4-gallon jug**
- One **3-gallon jug**
- No measuring marks
- Unlimited water from a pump

Goal: Obtain exactly **2 gallons in the 4-gallon jug**.

Solution

Represent a state as **(x, y)**, where:

- x = amount of water in 4-gallon jug
- y = amount of water in 3-gallon jug

Initial state:

(0, 0)

Goal state:

(2, 0)

Steps

Step	Operation	State (4G, 3G)
1	Start with both jugs empty	(0,0)
2	Fill the 3-gallon jug	(0,3)
3	Pour 3G into 4G	(3,0)
4	Fill the 3-gallon jug again	(3,3)
5	Pour 3G into 4G until 4G is full	(4,2)
6	Empty the 4-gallon jug	(0,2)
7	Pour the 2 gallons from 3G into 4G	(2,0)

Final Answer

Therefore, exactly **2 gallons of water are obtained in the 4-gallon jug.**

Goal state = (2,0).

2. Mars Rover as an Intelligent Agent

A Mars Rover is an intelligent agent that observes the Martian environment, makes decisions, moves around, and collects/analyses samples. The question asks about its precepts, environment, actions, performance and suitable architecture.

i. Precepts

The rover receives information from its sensors, such as:

- Images of the Martian surface
- Distance to obstacles
- Terrain condition
- Temperature
- Atmospheric conditions
- Chemical/mineral composition of samples
- Position and orientation
- Battery/energy level

These precepts help the rover understand its surroundings.

ii. Operating Environment

The Mars Rover operates in an environment that is:

- **Partially observable** – it cannot observe the entire planet.
- **Dynamic** – environmental conditions can change.
- **Unknown** – the rover does not know everything about its surroundings beforehand.
- **Sequential** – present actions affect future situations.
- **Continuous** – movement and measurements occur continuously.
- **Single-agent** – generally considered to operate independently.

iii. Actions

The rover can:

1. Move forward/backward.
2. Turn left or right.
3. Avoid obstacles.
4. Navigate toward a target.
5. Collect soil and rock samples.
6. Analyse samples.
7. Capture photographs.
8. Communicate information to Earth.
9. Adjust its position and orientation.

iv. Performance Measure

The rover's performance can be evaluated using:

- Number of useful samples collected and analysed.
- Accuracy of scientific measurements.
- Distance successfully explored.
- Successful navigation without damage.
- Energy efficiency.
- Amount and quality of useful data transmitted to Earth.
- Successful completion of assigned missions.

v. Suitable Agent Architecture

A **model-based, goal-based intelligent agent** is suitable for a Mars Rover.

Reason:

The rover cannot directly observe the entire environment. Therefore, it needs an **internal model** of its surroundings. It also has specific goals such as:

- Reaching a particular location.
- Avoiding obstacles.
- Collecting samples.
- Analysing samples.

Thus, a **model-based goal-oriented architecture** allows the rover to maintain information about its environment and select actions that help achieve its goals.

Conclusion

The Mars Rover is an intelligent agent because it **perceives its environment through sensors, makes decisions, performs actions through actuators, and evaluates its performance based on mission goals.**

3. 8-Queens Problem – Search Space and Problem Formulation

The **8-Queens problem** requires placing **8 queens on an 8×8 chessboard** such that no two queens attack each other. A configuration is not a solution if any two queens are in the same row, column or diagonal.

Problem Formulation

1. Initial State

The chessboard is initially **empty**.

[]

No queens have been placed.

2. State

A state represents a **partial placement of queens** on the chessboard.

For example:

[1, 3, 5]

can represent queens placed in the first three columns at rows 1, 3 and 5.

3. Actions

Place a queen in the next column such that it does not attack any previously placed queen.

A queen must not share:

- Same row
- Same column
- Same diagonal

4. Transition Model

When a valid position is selected, a queen is placed in the next column.

For example:

$$[1] \rightarrow [1, 3] \rightarrow [1, 3, 5]$$

Each transition adds one valid queen.

5. Goal Test

The goal is reached when:

8 queens are placed

and no two queens attack each other.

One valid solution is:

$$[1, 5, 8, 6, 3, 7, 2, 4]$$

This means:

- Column 1 $\rightarrow$ Row 1
- Column 2 $\rightarrow$ Row 5
- Column 3 $\rightarrow$ Row 8
- Column 4 $\rightarrow$ Row 6
- Column 5 $\rightarrow$ Row 3
- Column 6 $\rightarrow$ Row 7
- Column 7 $\rightarrow$ Row 2
- Column 8 $\rightarrow$ Row 4

6. Path Cost

For the basic 8-Queens problem, every queen placement can be considered to have equal cost.

Therefore:

$$\text{Path Cost} = 8$$

for a complete solution.

Search Space

The search starts from the empty board and generates valid partial configurations:

$$\text{Empty board} \rightarrow 1 \text{ queen} \rightarrow 2 \text{ queens} \rightarrow \dots \rightarrow 8 \text{ queens}$$

Invalid states are eliminated whenever queens attack each other.

Conclusion

The 8-Queens problem can be solved using search techniques such as **Depth-First Search, Backtracking, Breadth-First Search**, etc. Backtracking is particularly effective because invalid configurations can be rejected early.

4. OLA Cab Booking – Problem-Solving Agent

The question describes a customer travelling from one location to another using an OLA-like cab booking application, where different cab types can be selected.

Type of Problem-Solving Agent

A **goal-based problem-solving agent** is suitable.

Goal

The main goal is:

Travel from the source location to the destination safely and comfortably.

The agent determines the possible cab choices and selects one based on the customer's requirements.

Problem Formulation

Initialstate:

Customer is at the source location.

Goalstate:

Customer reaches the destination.

Actions:

- Select cab type
- Enter destination
- Book cab
- Travel through available route
- Reach destination

Statespace:

Different cab types and possible routes between source and destination.

Pathcost:

May include fare, travel time, distance and other factors.

Pseudocode

BEGIN

Input source location

Input destination location

Display available cab types

Mini

Micro

Sedan

Shared

Prime

Get customer's preference

For each available cab:

Calculate estimated fare

Calculate estimated travel time

Check availability

Select the cab satisfying customer's requirements

Book the selected cab

Find suitable route from source to destination

While destination is not reached:

Follow the selected route

Update current location

Reach destination

Display trip details and fare

END

Conclusion

A **goal-based agent** is appropriate because it selects actions based on the goal of **reaching the destination according to the customer's preferences**.

5. Uniform Cost Search – Logistics Delivery Network

The question asks us to find the **least-cost path from S to G using Uniform Cost Search (UCS)**. The given graph has the following weighted paths.

From the diagram:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Uniform Cost Search

UCS always expands the node having the **lowest cumulative path cost**.

Step 1: Start at S

S

Cost:

$$g(S) = 0$$

Successors:

$$A: 1, G: 12$$

Choose **A**, because $1 < 12$.

Step 2: Expand A

From A:

$$A \rightarrow B = 3$$

Therefore:

$$g(B) = 1 + 3 = 4$$

and

$$A \rightarrow C = 1$$

Therefore:

$$g(C) = 1 + 1 = 2$$

Frontier:

Node	Cost
C	2
B	4
G	12

Choose **C**.

Step 3: Expand C

From C:

$$C \rightarrow D = 1$$

So,

$$g(D) = 2 + 1 = 3$$

Also:

$$C \rightarrow G = 2$$

So,

$$g(G) = 2 + 2 = 4$$

Frontier:

Node	Cost
D	3
B	4
G	4

Choose **D**.

Step 4: Expand D

From D:

$$D \rightarrow G = 3$$

Cost:

$$g(G) = 3 + 3 = 6$$

But an existing path to G has cost **4**, so we retain the cheaper path.

Step 5: Select G

The lowest-cost goal node is:

$$G = 4$$

Therefore, UCS terminates.

Final Path

$$\boxed{S \rightarrow A \rightarrow C \rightarrow G}$$

Total Cost

$$1 + 1 + 2 = 4$$

$$\boxed{\text{Minimum Cost} = 4}$$

Comparison

Direct path:

$$S \rightarrow G = 12$$

Other path:

$$\begin{aligned} S &\rightarrow A \rightarrow B \rightarrow D \rightarrow G \\ &= 1 + 3 + 3 + 3 = 10 \end{aligned}$$

Our UCS path:

$$\begin{aligned} S &\rightarrow A \rightarrow C \rightarrow G \\ &= 1 + 1 + 2 = \boxed{4} \end{aligned}$$

Conclusion

Using **Uniform Cost Search**, the least-cost delivery route is:

$$\boxed{S \rightarrow A \rightarrow C \rightarrow G}$$

with a minimum total cost of:

$$\boxed{4}$$